OLIST MARKETPLACE ANALYSIS

What's Really Driving Customer Satisfaction on Brazil's Largest Marketplace Integrator

Data Analytics Hackathon — Gradient Learnings
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Dataset: Olist Brazilian E-Commerce Public Dataset (Sep 2016 – Oct 2018)

Executive Summary

96,478	**4.07 / 5**	**6.8%**	**1.69 / 5**
Delivered orders analyzed	Platform average review	Orders delivered late	Avg. review when 7+ days late

- Satisfaction tracks one thing above all: whether the order arrived on time. Not price, category, or payment method.
- 1 day late → avg. review falls 4.28 → 2.70. 7+ days late → falls to 1.69. Confirmed via logistic regression, controlling for price, basket size, and payment terms.
- Second strongest driver: orders split across multiple sellers are 6.5x more likely to get a bad review — even when delivered on time. Points to a coordination problem, not a speed problem.
- Root cause is structural: sellers are concentrated in São Paulo (60%); customers are more spread out. Cross-state orders take 2x longer and cost 75% more in freight. Worst-hit: AL, MA, SE, CE, PI (14–21% late rate).
- Every claim in this report was stress-tested — multicollinearity checked, model validated on held-out data, category rankings confirmed with confidence intervals. Where something didn't hold up, we say so.

1. Problem Understanding & Approach

1.1 Business Context

Olist connects small and medium Brazilian merchants to major online marketplaces through a single contract; sellers fulfill orders through Olist's logistics partners, and customers receive a post-delivery satisfaction survey (1–5 stars, optional comments). Leadership has ~2 years of order history but no consolidated view connecting delivery performance, geography, category, payment behavior, and satisfaction. This analysis builds that view and identifies where to focus improvement efforts as the platform scales.

1.2 Analytical Approach

- Consolidated 9 raw tables (99,441 orders; 112,650 line items; 103,886 payments; 100,000 reviews; 32,951 products; 3,095 sellers; ~1M geolocation pings) into an order-level master table and an item-level table for category/seller cuts.
- Cleaned data deliberately rather than blindly: non-delivered orders (3.5% of the dataset) were excluded from delivery-timing analysis since they lack completion timestamps by nature; multi-item and multi-payment orders were aggregated (summed) to the order grain rather than left to inflate row counts; geolocation was aggregated to one mean lat/lng per zip prefix before any geographic join.
- Used customer_unique_id (not customer_id) for anything about repeat purchase behavior, per the documented distinction between the two identifiers.
- Combined descriptive/exploratory analysis (trends, cross-tabs, visualizations) with a multivariate logistic regression to separate primary drivers of low review scores from factors that are merely correlated with them.

AI disclosure: I used Claude AI throughout this project — for writing and debugging analysis code, structuring the notebook/report, and drafting narrative text — while I directed the analytical approach and reviewed the findings. Per the hackathon's disclosure requirement.

2. Marketplace Performance Over Time

Monthly order volume grew from roughly 1,330 orders/month in the platform's early months to over 6,600/month by mid-2018 — a ~5x increase — while revenue tracked orders closely. Review scores, however, tell a more complicated story: they held in the 4.0–4.2 range through most of 2017, dipped notably in November 2017 (3.90) and again in February–March 2018 (3.81 / 3.74), before recovering to the platform's best-ever levels (4.25+) by mid-2018. We verified the November dip against Brazil's actual Black Friday date (confirmed: November 24, 2017) rather than assuming the coincidence — order volume and the review dip both center on that week, consistent with operational strain during the demand spike.

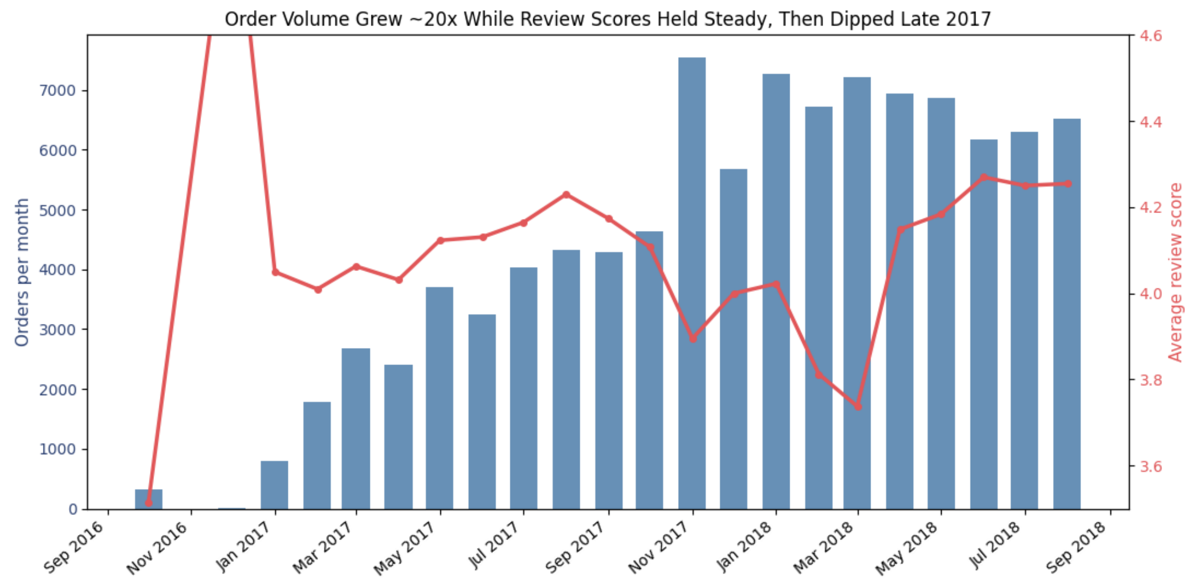

Figure 1. Monthly order volume (bars) vs. average review score (line). Review dips cluster around demand spikes and early 2018.

> **Insight**
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> Growth and satisfaction do not move in lockstep. The two dips that stand out — the Nov 2017 peak-demand month and the Feb–Mar 2018 stretch — are exactly the periods worth cross-checking against operational strain (see Section 3 on delivery).

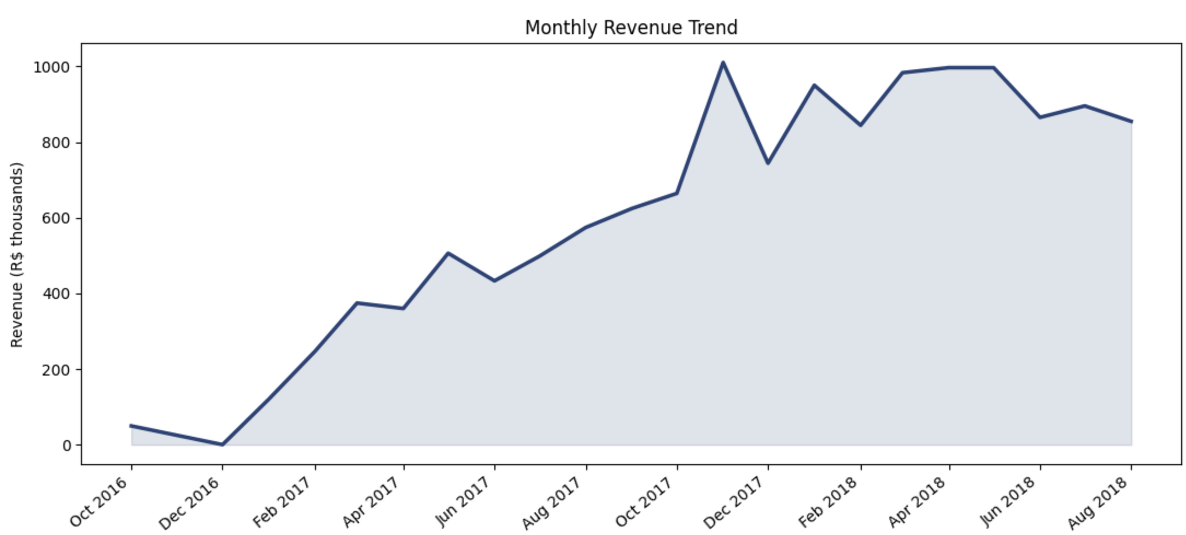


Figure 2. Monthly revenue trend, R$ thousands.

3. Delivery Performance and Customer Satisfaction

Across 96,470 delivered orders with complete timing data, 6.8% arrived after their estimated delivery date. The relationship between delay and satisfaction is the strongest single pattern in this entire dataset:

Delivery outcome	Avg. review score	Orders (n)
Very early (7+ days ahead)	4.30	75,772
Early (1–6 days ahead)	4.17	12,454
On time	4.02	1,279
Late (1–7 days)	2.70	3,652
Very late (7+ days)	1.69	2,844

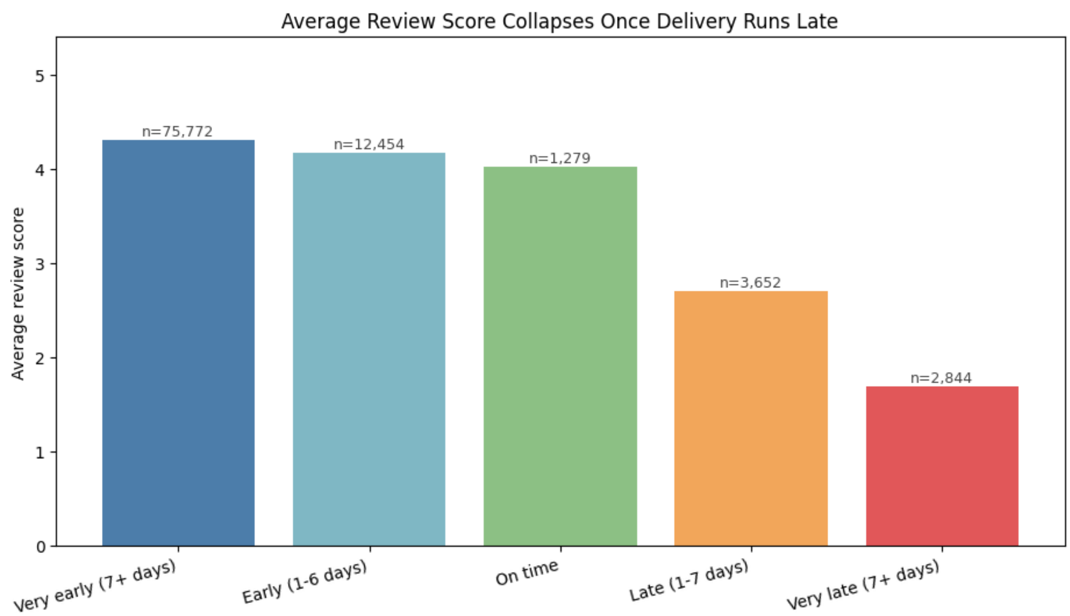


Figure 3. Average review score by delivery-timing bucket. The drop-off from 'on time' to 'late' is not gradual — it is a cliff.

Insight

This is not a gentle correlation (Pearson r = –0.27 across the full delay range) — it is a threshold effect. Being early costs Olist almost nothing in satisfaction; crossing the promised date at all drops average score by more than a full star, and by 7+ days late, a majority of customers are leaving 1-star reviews. Because Olist already sets the estimated delivery date, the highest-leverage fix is tightening estimate accuracy and carrier reliability, not just 'shipping faster' in the abstract.

4. Seller and Geographic Patterns

Sellers are far more geographically concentrated than customers: 59.7% of sellers are based in São Paulo state versus 42.0% of customers, with Paraná (11.3%) and Minas Gerais (7.9%) a distant second and third for sellers. This mismatch means a large share of orders must travel cross-state.

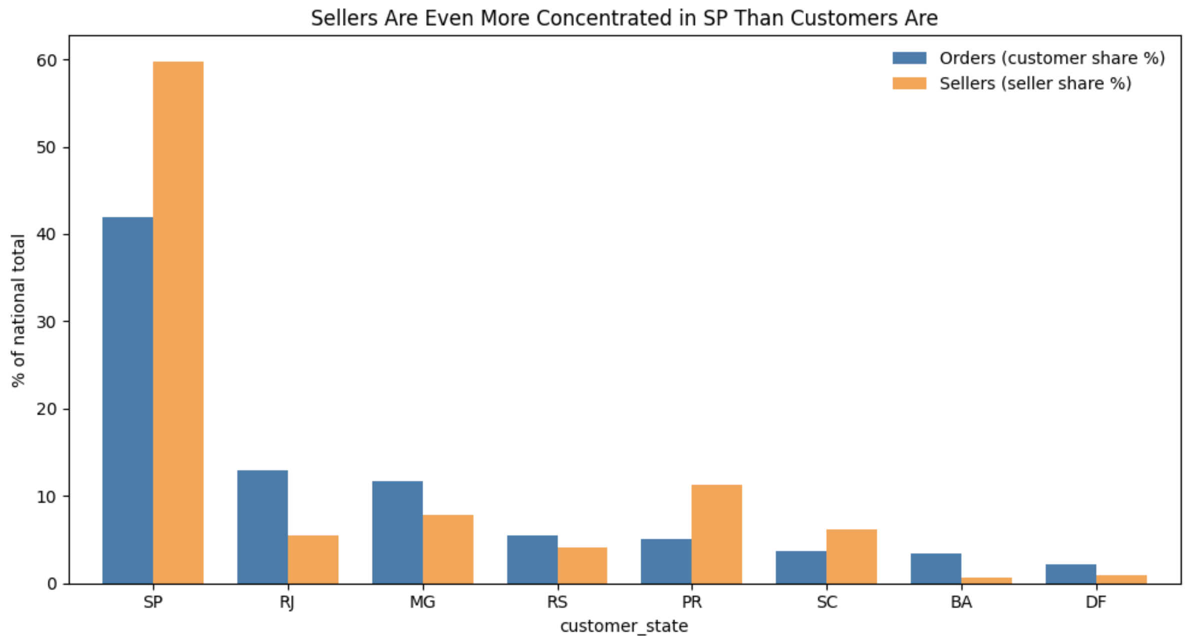
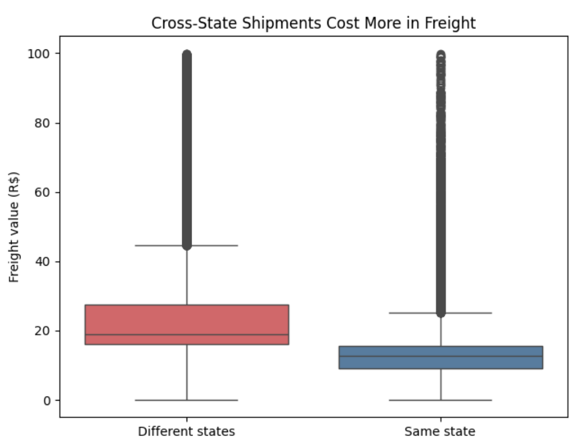


Figure 4. Seller concentration in the Southeast outpaces customer concentration in the same states.

Cross-state orders take nearly twice as long to arrive (14.7 days average vs. 7.5 days for same-state) and cost 75% more in freight (R$26.96 vs. R$15.46 average). This geographic mismatch is a structural, upstream driver of the delay problem identified in Section 3.



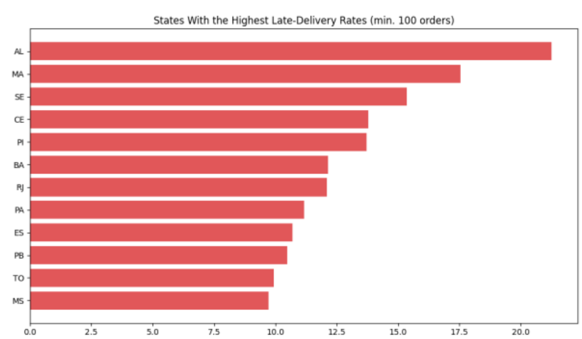


Figure 5 (left). Freight cost is materially higher for cross-state shipments. Figure 6 (right). A handful of states — Alagoas, Maranhão, Sergipe, Ceará, Piauí — see late-delivery rates 2–3x the platform average.

Insight

The states with the worst late-delivery rates (AL: 21%, MA: 18%, SE: 15%) are all in Brazil's North/Northeast — far from the Southeast seller base. This is a logistics-network gap, not a demand-quality issue: these regions are simply underserved by nearby fulfillment capacity.

5. Product Category Performance

Bed & Bath, Health & Beauty, and Sports & Leisure are the platform's three highest-volume categories. Review scores vary meaningfully by category even among high-volume ones: Office Furniture (3.51), Furniture Living Room (3.91), and Bed & Bath & Table (3.91) sit well below the platform average, while Books, Food & Drink, and Luggage & Accessories consistently score above 4.3.

With some category sample sizes as low as 80 orders, a raw ranking risks mistaking noise for a real pattern. We tested every category with a 95% confidence interval on its average review score against the platform-wide average: 12 of 54 tested categories are statistically — not just numerically — below average, and every furniture-adjacent category we flagged (Office Furniture, Furniture Living Room, Furniture Décor, Bed/Bath/Table, Home Comfort) holds up under this stricter test.

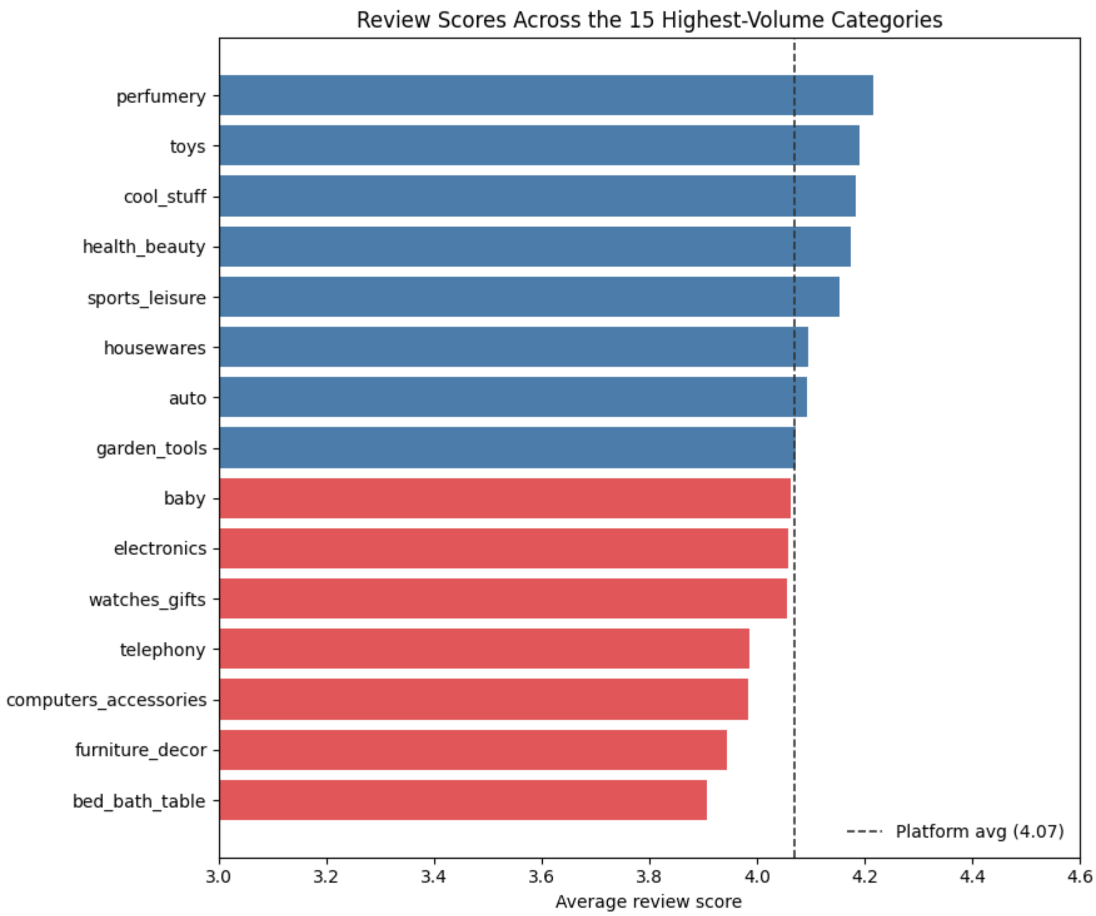


Figure 7. Average review score for the 15 highest-volume categories, relative to the platform average (dashed line).

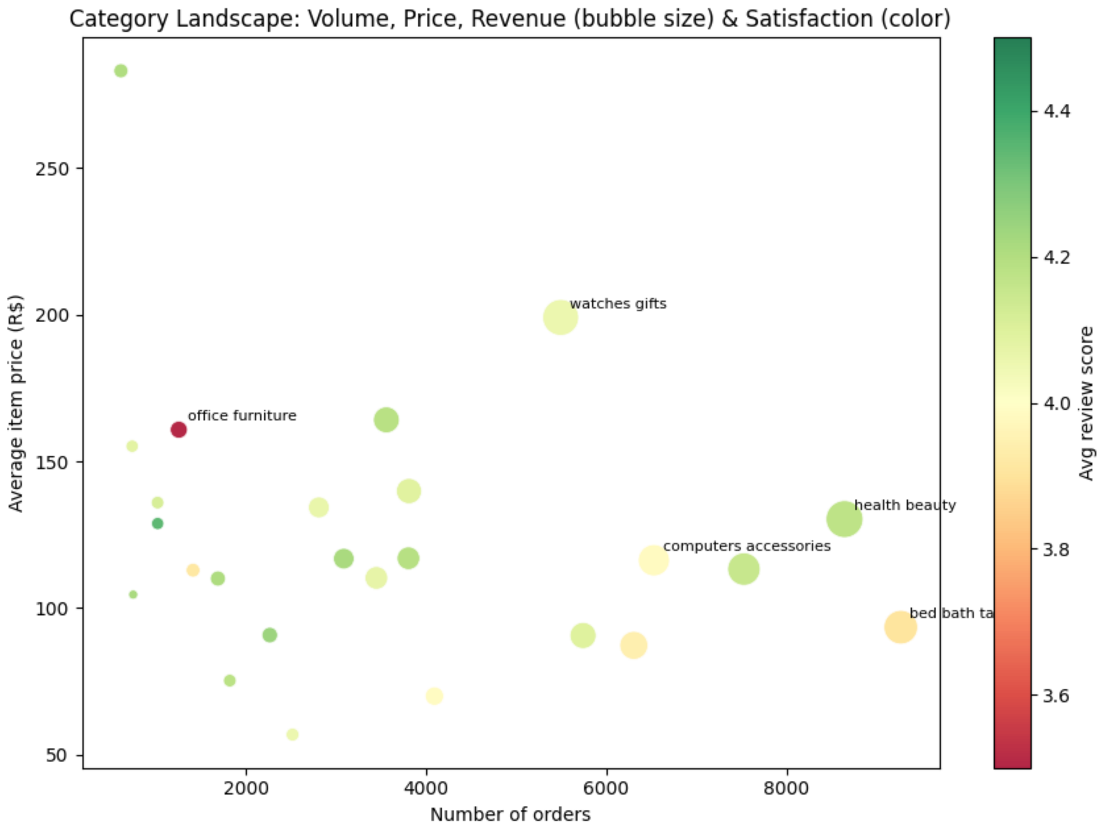


Figure 8. Category landscape — volume (x), average price (y), revenue (bubble size), and satisfaction (color). Furniture and office-furniture categories cluster on the lower end of satisfaction despite decent volume.

Insight

Furniture-adjacent categories (bed/bath/table, furniture décor, furniture living room, office furniture) underperform as a group, and this is confirmed rather than assumed. These are bulky, higher-damage-risk, harder-to-ship products — consistent with delivery friction rather than product quality being the driver, and worth cross-referencing against packaging/logistics partners specific to these categories.

6. Payment Behavior

Credit card is the dominant payment method (75.4% of orders), followed by boleto (19.9%). Average order value rises with credit card use (R$167) versus boleto (R$145), debit card (R$143), and vouchers (R$120, typically used alongside another method for smaller top-ups).

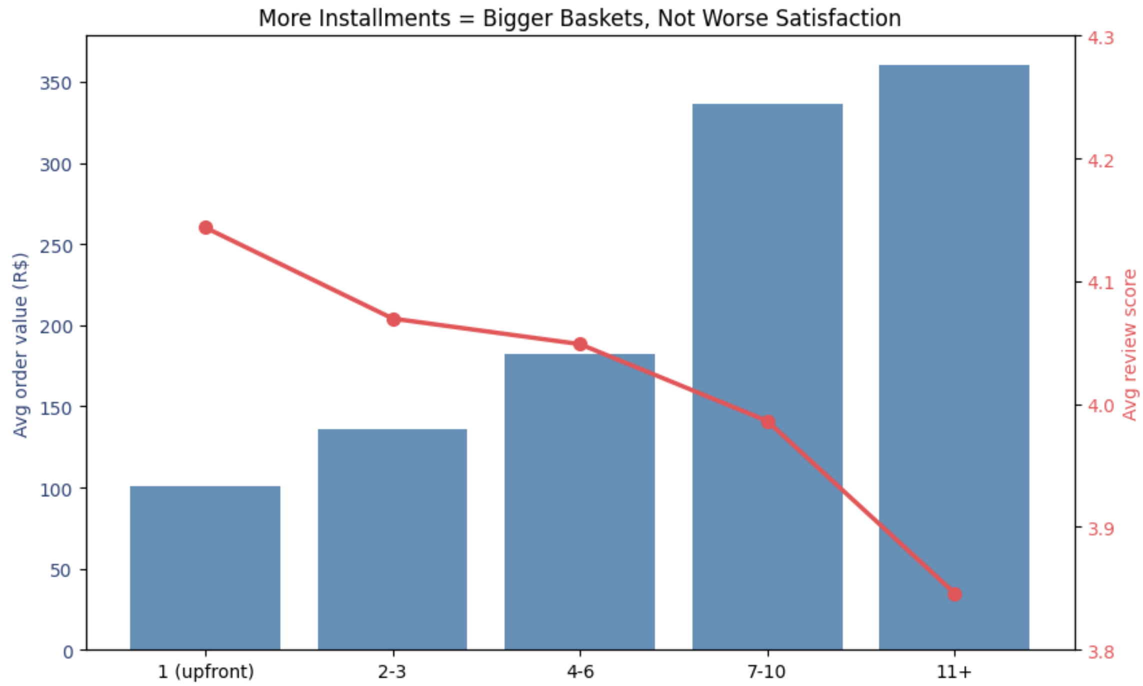


Figure 9. More installments correlate with larger baskets (bars) without materially hurting satisfaction (line) — until 11+ installments, a small, high-value segment worth watching.

Insight

Payment type and installment count are not meaningful drivers of dissatisfaction on their own — review scores stay within a narrow 3.85–4.23 band across every payment segment. Debit card users are both the smallest group and the most satisfied (4.23 avg, 5.3% late rate), suggesting no urgent payment-related friction to fix. This is a useful negative result: it lets Olist rule out payments as a satisfaction lever and focus resources on delivery instead.

7. Root-Cause Analysis: What Actually Drives Low Review Scores

To separate genuine drivers from factors that are merely correlated with each other, we fit a logistic regression predicting the odds of a 1– or 2–star review (13.1% of delivered orders) using delivery delay, order size, freight-to-price ratio, whether an order involved multiple sellers, order value, and payment installments simultaneously.

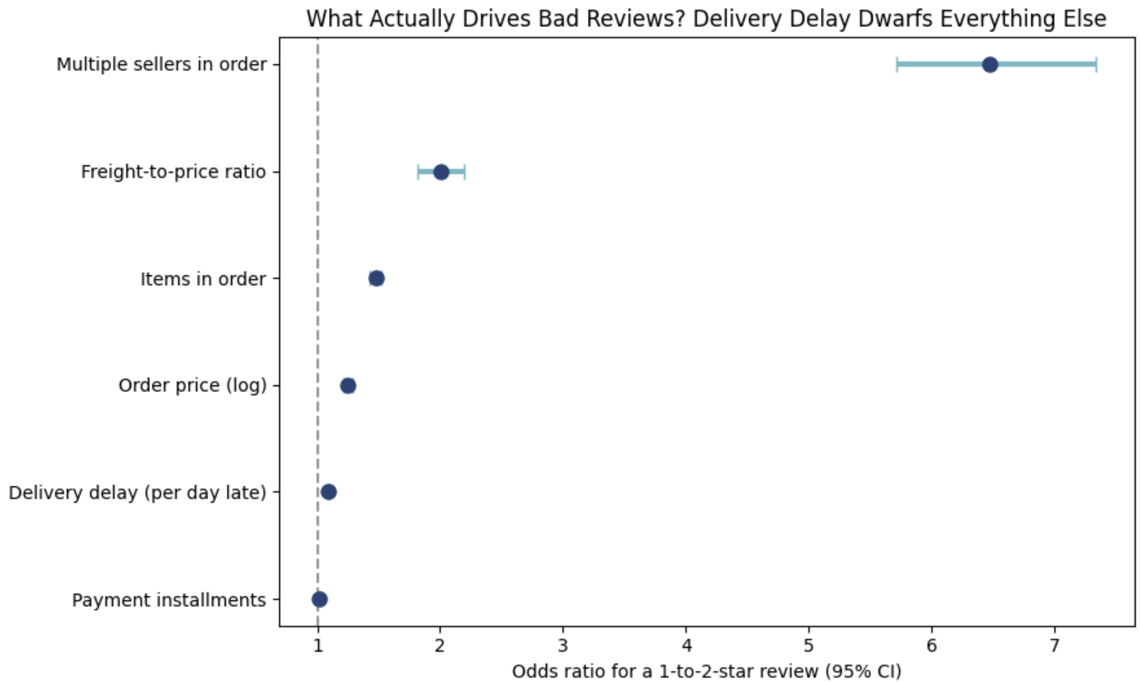


Figure 10. Odds ratios (95% CI) for a 1–2 star review. Bars crossing 1.0 have no effect; further right means a stronger driver of bad reviews.

Factor	Odds ratio	Interpretation
Multiple sellers in one order	6.47x	Primary driver — rare (1.3% of orders) but by far the strongest single effect, even controlling for delay
Delivery delay (per day late)	1.09x per day	Primary driver — compounds quickly: ~9% higher odds for every additional day late
Freight-to-price ratio	2.00x (high vs. low)	Secondary driver — orders where shipping cost is disproportionate to item price see more complaints
Order value (log price)	1.25x	Secondary — higher-value orders carry modestly higher expectations
Items per order	1.47x	Secondary — more items means more can go wrong
Payment installments	1.02x	Negligible — confirms Section 6's finding

Bottom line

Two primary drivers explain most of what's controllable here: (1) delivery delay, by far the highest-volume driver since it touches every late order, and (2) split-seller orders, a much rarer but disproportionately damaging failure mode — these orders are delivered slightly earlier on average, so the dissatisfaction is coming from coordination/communication friction (partial shipments, confusing tracking, mismatched arrival times) rather than raw speed. Freight-to-price ratio and order size are real but secondary contributors. Payment behavior is not a meaningful driver at all.

7.1 Robustness Checks

A regression result is only as trustworthy as the checks run against it. Three checks were run specifically against the two flagged primary drivers before including them here:

Check	Result
Multicollinearity (VIF)	All factors under 2.5 — no multicollinearity concern
Out-of-sample validation (75/25 split)	AUC = 0.70 on held-out data; multi-seller odds ratio nearly identical (6.66x vs. 6.47x) — not a fluke of one split
Confound isolation	Restricting to on-time, small-basket orders only: multi-seller orders still show a 46% low-review rate vs. 9% for single-seller — not explained by delay or order size

Being honest about the ceiling

The model's pseudo R^2 is 0.13 — these six features explain a real but modest share of what separates a bad review from a good one. Plenty of dissatisfaction is likely driven by things this model doesn't capture (product quality itself, individual seller behavior, expectations set by listing photos/descriptions). The multi-seller effect, while robust to every check we ran, is based on only 1,263 orders and still deserves a dedicated follow-up — we treat it as a strong lead, not a closed case.

8. What the Review Text Adds

58% of reviews have no written comment, but for the 41,753 that do, comparing word frequency between low-score (1–2 star) and high-score (4–5 star) reviews surfaces complaints that never show up in the structured columns.

Word (Portuguese)	Meaning	Ratio vs. high-score reviews
decepção / decepcionada	disappointment	127x / 30x
descaso	neglect, carelessness	119x
enganosa	misleading, deceptive	112x
falsificado	counterfeit	57x
reembolso / estorno	refund / chargeback	55x / 40x
atrasada	delayed	45x
procon	Brazil's consumer-protection agency	27x

Insight

Beyond delay, two extra clusters emerge from customers' own words: demands for refunds or chargebacks, and — more concerning for platform trust — complaints about counterfeit or misrepresented products (‘enganosa’, ‘falsificado’). Mentions of PROCON signal customers escalating to formal consumer complaints, a reputational and regulatory signal worth monitoring on its own. This is a simple word-frequency pass, not a full NLP model, but it's a real, cheap signal the numeric score alone misses — worth a deeper look at which sellers or categories these complaints cluster around.

9. Additional Finding: Repeat Customers

Using the person-level customer_unique_id (not the order-level customer_id), only 3.1% of Olist's ~96,000 unique customers placed more than one order, contributing just 5.7% of total revenue. Repeat customers rate their experience marginally higher on average (4.15 vs. 4.07), but the platform is overwhelmingly a one-time-purchase marketplace today — a retention-side opportunity that sits outside this report's primary satisfaction focus but is worth flagging for leadership.

10. Recommendations, Prioritized

Priority 1 — Fix delivery estimate accuracy and Northeast/North logistics coverage

- Tighten estimated-delivery-date calculation for cross-state and Northeast-bound shipments specifically — the data shows Olist is systematically over-promising in exactly the regions least able to deliver on that promise.
- Investigate regional fulfillment or logistics-partner options for the five worst-performing states (AL, MA, SE, CE, PI) to shrink the cross-state distance penalty.

Priority 2 — Fix multi-seller order coordination

- Even though rare (1.3% of orders), split-seller orders are the single strongest predictor of a bad review (6.5x). Audit the multi-seller checkout and tracking experience — likely candidates are confusing partial-shipment notifications and mismatched delivery windows across sellers.

Priority 3 — Targeted support for underperforming categories

- Bed & Bath, Furniture Décor, Furniture Living Room, and Office Furniture score meaningfully below platform average despite solid volume — review packaging standards and preferred-carrier assignment for bulky/fragile categories specifically.

Lower priority / monitor only

- Payment type and installment count show no meaningful link to satisfaction — no action needed here beyond continued monitoring.
- Repeat-purchase rate (3.1%) is low; a retention/loyalty initiative is a plausible longer-term growth lever, though it is a separate workstream from the satisfaction question this report focuses on.

11. Limitations

- This is observational data; regression results identify strong, validated associations — not proven causation, even after the robustness checks in Section 7.1.

- The multi-seller effect has a small sample (1,263 orders). We ruled out delay and basket size as explanations, but not seller identity or full category mix — it merits a dedicated follow-up study before an operational change is built on it alone.
- Review text analysis (Section 8) used simple word-frequency comparison on the 42% of reviews with comments — a real signal, but a proper topic or sentiment model would extract considerably more, including whether counterfeit complaints cluster around specific sellers.
- Geolocation was aggregated to a zip-prefix mean lat/lng, which is a reasonable approximation but not exact for precise distance calculations.